Lecture 3: Neural Networks



Course: Deep Learning Fundamentals

1. Introduction

A neural network is a computational model inspired by the structure of biological brains. It consists of layers of interconnected neurons that propagate signals.

Key topics covered today:
- Forward propagation and activations
- Loss functions such as cross-entropy
- Gradient descent with learning rate alpha

The update rule is:

$$\theta = \theta - \alpha \nabla J(\theta)$$

2. Comparison of activation functions

Activation	Range	Use case
ReLU	0 to inf	Hidden layers
Sigmoid	0 to 1	Binary output
Softmax	0 to 1	Multi-class

2. Section content

Paragraph 1 on page 2 covers training dynamics, regularization and evaluation metrics.

Paragraph 2 on page 2 covers training dynamics, regularization and evaluation metrics.

Paragraph 3 on page 2 covers training dynamics, regularization and evaluation metrics.

Paragraph 4 on page 2 covers training dynamics, regularization and evaluation metrics.

3. Section content

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